PART II: PATTERNS

Topic 13: Smoothing and Moving Averages

The Core Concept

Smoothing reduces the noise in data to reveal the underlying signal. The most common tool is the **moving average**, which replaces each data point with the average of itself and a set number of preceding points.

Why It Matters: Finding the Signal

When data is noisy, month-to-month changes can feel alarming even when nothing is really changing. A 3-month moving average helps answer: "Are we genuinely trending in a new direction, or is this just random variation?"

The Example

Note: All data in this tutorial series are synthetic, created for teaching purposes only. They do not represent real-world incidents or any specific jurisdiction.

Month	Raw Count (N. North)	3-Month Moving Avg	Interpretation
Jan	12	12.0	Baseline (only 1 month)
Feb	10	11.0	Slight dip — or just noise?
Mar	14	12.0	"Spike" smoothed away
Apr	15	13.0	Gradual upward movement
May	18	15.7	Trend becoming visible
Jun	22	18.3	Clear upward signal
Jul	25	21.7	Summer peak confirmed
Sep	19	—	Aug missing; avg skipped
Oct	16	17.3	Post-summer decline
Nov	14	15.7	Continuing downward
Dec	11	13.7	Returning to winter baseline

The raw count for March (14) looks like a 40% jump from February (10) — potentially alarming. But the 3-month moving average for March is 12.0, exactly where we started in January. The apparent spike was just the data returning to its mean after a quiet February.

Note: When August data is missing, the moving average for that month cannot be calculated — another reason missing data is problematic (see Topic 6).

💡 **The Insight:** The smoothed line tells you where the data is actually heading. The raw line tells you what happened on any given month.

Key Takeaway: Use a moving average when you want to answer "Are we better off now than we were three months ago?" — not "What happened last Tuesday?"